

Preparation and Thermochromic Switching between Phosphorescence and Thermally Activated Delayed Fluorescence of Mononuclear Copper(I) complexes

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General Information

All chemicals were purchased from commercial sources (Acros Organics, Sigma-Aldrich, Alfa Aesar) and were used without further purification.

DFT calculations were carried out using the ORCA program package (version 4.0.1).¹ All calculations were performed using the B3LYP functional^{2–4} and employ the RIJCOSX approximation.^{5,6} Relativistic effects were calculated at the zeroth order regular approximation (ZORA) level.⁷ The ZORA keyword automatically invokes relativistically adjusted basis sets. To account for solvent effects, a conductor-like polarizable continuum model (CPCM) modeling dichloromethane was used in all calculations.⁸ Geometry optimizations were performed using Ahlrichs' split-valence triple- ξ basis set def2-TZVP, which comprises polarization functions for all non-hydrogen atoms.^{9,10} Atom-pairwise dispersion correction was performed with the Becke-Johnson damping scheme (D3BJ).^{11,12} The presence of energy minima was checked by numerical frequency calculations. TD-DFT calculations with 50 vertical transitions were carried out based on def2-TZVP optimized geometry of the respective complex. Explicit solvent molecules were not taken into account.

Table S1. Photophysical data (emission wavelength, emission lifetime, emission quantum yield and singlet triplet energy difference) of CuX(PPh₃)₂L complexes in the solid state at 298 K and 77 K (in parentheses).

complex	λ_{em} [nm]	τ_{em} [μ s]	ϕ_{em}	ΔE (¹ MLCT– ³ MLCT) [cm ^{−1}]
CuCl(PPh ₃) ₂ (py) ¹³	498 (516)	12 (27)	0.98 (0.89)	740
CuBr(PPh ₃) ₂ (py) ¹³	483 (496)	22 (49)	0.95 (0.93)	850
CuI(PPh ₃) ₂ (py) ^{13,14}	485 (506)	19 (91)	0.99 (0.99)	810
CuCl(PPh ₃) ₂ (py ^{Me}) ¹⁵	468 (488)	9.4 (36)	0.99 (0.82)	940
CuBr(PPh ₃) ₂ (py ^{Me}) ¹⁵	467 (477)	15 (52)	0.95 (0.95)	1070
CuI(PPh ₃) ₂ (py ^{Me}) ¹⁵	455 (458)	9.5 (52)	0.66 (0.76)	1170
CuI(PPh ₃) ₂ (py ^{tBu}) ¹⁶	454 (470)	(21.6)	1.00	
CuCl(PPh ₃) ₂ (pym) ¹⁷	526	3.8	0.40	
CuCl(PPh ₃) ₂ (pyz ^{Me}) ¹⁷	627	3.7	0.09	
CuCl(PPh ₃) ₂ (iq) ¹³	491, 515, 553 (476, 488, 511, 522, 549)	506 (1285)	0.77 (0.56)	
CuCl(PPh ₃) ₂ (nap) ¹³	636 (674)	1.5 (14)	0.16 (0.08)	

Instructions for the students (experimental section)

1. Equipment

Mortar with pestle, spatula, Büchner funnel, filter paper, filtering flask, sleeve for vacuum filtration, tweezer, UV hand lamp, vials, Dewar flask

2. Hazards and Disposal

Before carrying out the experiments, the safety data sheets of all chemicals used have to be read and understood.

Pyridine and many substituted pyridines are (very) toxic. Pyridine is a possible reproductive toxin, it easily dissolves in water and harms both animals and plants in aquatic systems, and it is harmful if inhaled, swallowed, or absorbed through the skin. Triphenylphosphane irritates the skin, eyes, and respiratory tract and is harmful if swallowed or inhaled. Copper(I) halides are hazardous in case of ingestion and inhalation, and irritate the skin, eyes, and respiratory tract. They are also dangerous for the environment. Pyridine and *tert*-butyl methyl ether are flammable. Dichloromethane may cause cancer and damage organs through prolonged or repeated exposure. Liquid nitrogen causes cold burnings, hence wearing cryo gloves is recommended. For glass Dewar flasks, the danger of implosion is given. All experiments should be performed in a fume hood. Care should be taken to avoid contact during the preparation. Gloves and safety glasses have to be worn. Prevent irradiation of skin or eyes with the UV hand lamp (standard TLC viewing lamp) by turning-on the lamp only in the almost closed fume hood, wearing gloves and using a tweezer to irradiate the filter papers.

Collect solutions in the waste containers provided.

3. Preparation of mononuclear copper(I) complexes $\text{Cu}(\text{PPh}_3)_2(\text{py}^{\text{R}})$

3.1 Synthesis of $\text{Cu}(\text{PPh}_3)_2(\text{py})$ 1 ¹⁴

In a mortar, grind copper(I) iodide (0.20 g, 1.1 mmol) using the pestle in the presence of pyridine (1.6 mL) which serves as ligand and solvent for one minute. Observe the mixture under UV irradiation (365 nm) with a UV hand lamp in a dimmed fume hood or alternatively in a cardboard box. Add triphenylphosphane (0.80 g, 3.1 mmol) and grind the mixture for a few minutes. Observe the mixture under UV irradiation (365 nm) with a UV hand lamp in a dimmed fume hood again. Add *tert*-butyl methyl ether (5 mL) and collect the colorless precipitate by filtration on a filter paper. Wash the product several times with small portions of *tert*-butyl methyl ether, dry the complex in air and collect the product in a vial. Keep the filter paper.

3.2 Synthesis of $\text{Cu}(\text{PPh}_3)_2(\text{py}^{\text{CN}})$ 2

In a mortar, grind copper(I) iodide (0.20 g, 1.1 mmol) using the pestle for one minute in the presence of 4-cyanopyridine (4-pyridinecarbonitrile) (0.80 g, 7.7 mmol) and a few drops of dichloromethane to obtain a pasty-like mixture. Observe the mixture under UV irradiation (365 nm) with a UV hand lamp in a dimmed fume hood or alternatively in a cardboard box. Add triphenylphosphane (0.80 g, 3.1 mmol) and grind the mixture for a few minutes. Observe the mixture under UV irradiation (365 nm) with a UV hand lamp in a dimmed fume hood again. Add *tert*-butyl methyl ether (5 mL) and collect the colorless precipitate by filtration on a filter paper. Wash the product several times with small portions of *tert*-butyl methyl ether, dry the complex in air and collect the product in a vial. Keep the filter paper.

4. Demonstration of the temperature dependent luminescence of **1** and **2**

Pick a filter paper with residual precipitate **1** or **2** using a long tweezer and observe the filter paper under UV light (365 nm) in a dimmed fume hood or alternatively in a cardboard box at room temperature. Dip the filter paper with the tweezer in liquid nitrogen in a Dewar flask and observe under UV light again.

5. Question to the students

- Why are heavier metals less abundant on the Earth than the first row transition metals?
- Give reasons why mechanochemical syntheses can qualify as sustainable (green) methods!
- Which structural type does solid CuI adopt at ambient conditions? Describe the coordination sphere around copper and iodide.
- Which coordination number and geometry do copper(I) ions prefer and why?
- Find analogies between the solid structure of CuI and the pyridine complexes $\text{CuI}(\text{py})_n$ ($n = 1-3$).
- Which electronic transitions are responsible for the observed emission phenomena?
- What type of luminescence occurs at room temperature (r.t.) and at 77 K, respectively?
- Why do complexes **1** and **2** show different emission wavelengths?
- What do the different emission colors at r.t. and at 77 K tell us with respect to the relative energies of the emissive states for **1** and **2**? Draw qualitative Jablonski diagrams for **1** and for **2**.
- Why does the geometry of the complexes **1** and **2** change in solution after irradiation?
- Why is the luminescence of **1** and **2** only observed in the solid state, but not in solution?
- $[\text{CuI}(\text{py})]_4$ even shows luminescence in solution. Give a possible explanation for the different behavior of $[\text{CuI}(\text{py})]_4$ and **1** or **2**.
- Suggest ligand modifications which could enable luminescence of mononuclear four-coordinate copper(I) complexes even in solution.

Instructor's notes

1. Students typically complete the syntheses (starting from CuI) and luminescence demonstrations within one hour. Consequently, the students can perform this experiment within a more lengthy synthetic procedure such as refluxing or crystallization periods.
2. Other substituted pyridines and nitrogen heterocycles have been used for the syntheses of analogous copper(I) complexes as well. However, most of them are toxic and/or expensive. In this student's laboratory experiment, inexpensive pyridine and 4-cyanopyridine serve to demonstrate the effect of electron withdrawing substituents at the N-donor ligand.
3. The initially formed pyridine complexes with formal stoichiometries $\text{CuI}(\text{py})_n$ ($n = 1-3$) are tetrameric, dimeric and monomeric, respectively (Figure 1).^{18,19} Analogous structural motifs are likely with the 4-pyridinecarbonitrile ligand py^{CN} .
4. The collected dried samples of **1** and **2** can be stored in separate stoppered vials. These materials are useful for impressive lecture presentations for larger student groups.
5. This experiment is perfectly suitable to demonstrate conceivable contaminations with **1** in the fume hood, on lab coats, gloves, glass ware, spatula, etc. simply with a UV lamp. The outcome of this demonstration can increase the student's awareness for careful and safe working in the laboratory.
6. To explain the thermal switching between thermally activated fluorescence (TADF) and phosphorescence, it has proven helpful to describe the phenomena in a reaction type scheme with respective microstates in the T_d symmetry notation for simplification (Figure 4; splitting of the d orbitals in e and t_2 levels). Depending on their individual level, this description is easier to grasp than energy state diagrams for some students. A reaction sequence can formulate the photophysical and subsequent processes, showing TADF at r.t. and phosphorescence at 77 K, respectively, according to Figures 4 and 5. The description using microstates clarifies that the preferred coordination geometry in the MLCT states ($d^9\text{-Cu}^{\text{II}}$) is square-planar. This flattening of the tetrahedral to a square-planar geometry is prevented in the solid state but possible in solution.
7. The Cartesian coordinates (x,y,z) provided below for the model copper complexes can be used to demonstrate interactive computer models of the ground state and the excited state geometry using suitable visualizing software or to create 3D models using a 3D printer.
8. Complexes **1** and **2** can be characterized by IR spectroscopy. Corresponding ATR-IR spectra of **1** and **2** are depicted in Figure S1. The observed IR absorption bands of **1** and **2** are very similar and only small differences are observed in the fingerprint region (Figure S1, Table S6). The expected band for the CN vibration of **2** is very weak as in the ligand 4-cyanopyridine itself. This very weak band is only observed in a concentrated KBr disk at around 2232 cm^{-1} .

Student teacher interaction and assessment

This experiment clearly focuses on the direct observation of different luminescence phenomena augmented with the theoretical background. The synthetic procedure itself is not very demanding on the practical skills of the students. The experiments are performed in small groups of two to four students; often a pair of students prepares complex **1** and the other pair prepared complex **2**. The students perform the irradiation experiments in the presence of a lab assistant, typically a PhD student. The visual experimental observations directly lead to a discussion of the observations within this small learning group. At this level, the students already have knowledge about basic crystal/ligand field theory, the Jahn-Teller theorem, electronic transitions after absorption of photons in transition metal complexes (MC/ligand field, MLCT, LMCT, MM'CT and LL'CT transitions), the spin selection rule, Boltzmann distribution, as well as fluorescence and phosphorescence phenomena.

To describe the nature of the involved electronic states of the copper complexes, orbital sketches in the one electron approximation are used (Figure 4). These orbital descriptions are transformed to an energy state diagram with the help of the lab assistant (Figure 4). From physical chemistry lectures, the students know that the triplet state is lower in energy than the singlet state. With these collected information, the students are able to assign the different emissions at high and low temperature based on the experimental observation, that the emission shifts bathochromically, i.e. to lower energy, at lower temperature. The lab assistant gives further detailed information about the individual photophysical processes (internal conversion IC, intersystem crossing ISC) occurring between absorption and emission. With the aid of the lab assistant, the group develops the relative populations of the singlet and triplet excited states with reference to the Boltzmann population (known from physical chemistry). Based on the spin selection rule, the group develops the relative lifetimes of the singlet and triplet excited states and the relative rates of the respective luminescence processes. A hint to organic chemistry courses and the known effects of electron-withdrawing/donating substituents help to identify the underlying reason of the different emission colors of **1** and **2**. The flattening distortion is illustrated and explained by the DFT calculated structures of $\text{Cu}(\text{PMe}_3)_2(\text{py})$, given in the student's lab course script (Figure 5b,c). In a group seminar accompanying the lab course, the experiments and observations are discussed a second time in learning groups of thirty to forty students and an instructor. The group seminar also addresses further advanced details, e.g. the Jahn-Teller distortion, the flattening distortion and possible solutions to this issue.

All students are requested to write lab reports on the experiments, the observations and the explanations. The lab assistant assesses these written reports. The students typically perform well in writing their lab reports on these experiments. An oral examination on the complete module assesses the learning outcome of each individual student. The entire module comprises two lab courses on inorganic chemistry accompanied with seminars and lectures. The oral examinations reveal that the students understood the experiments and conclusions of the described experiments particularly well and very positively remembered these experiments. When asked to name the most interesting experiment of the lab course, the students select this experiment high above average.

Table S2. CAS Registry Numbers for the chemicals employed.

Name	CAS registry number
Copper(I) iodide	7681-65-4
Pyridine	110-86-1
4-Cyanopyridine	100-48-1
Triphenylphosphane	603-35-0
<i>tert</i> -Butyl methyl ether	1634-04-4
Dichloromethane	75-09-2

Table S3. Crystallographic data of the pyridine intermediates and **1**.

Name	CCDC number	Reference
[CuI(py)] ₄	1132856	20
[CuI(py) ₂] ₂	1138434	21
1	1193315	22

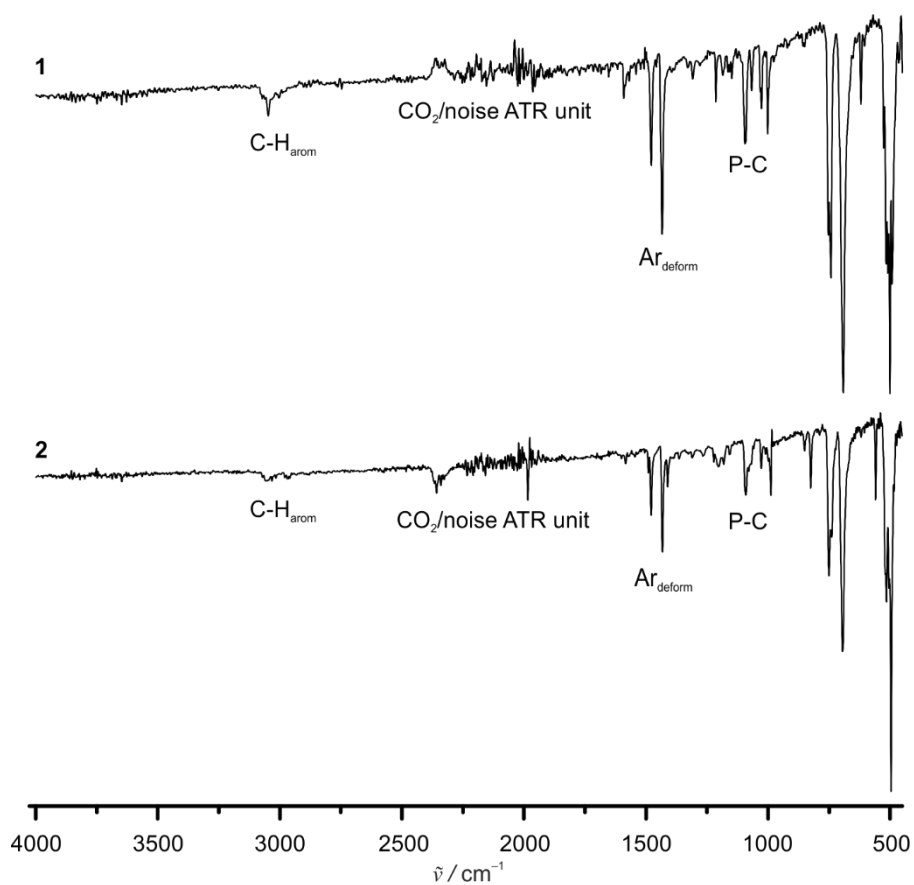


Figure S1. ATR IR spectra of **1** and **2**.

Table S4. DFT calculated Cartesian coordinates of geometry optimized $^1[\text{CuI}(\text{PMe}_3)_2(\text{py})]$

Cu	2.483310000	-0.301997000	-0.390093000
P	3.390574000	1.419909000	-1.485160000
P	2.057330000	-0.328538000	1.807352000
C	4.520387000	-4.561511000	-1.579915000
C	5.223811000	-3.577623000	-0.895085000
C	4.586514000	-2.381930000	-0.606073000
N	3.317930000	-2.131734000	-0.958011000
C	2.643455000	-3.080222000	-1.618351000
C	3.206203000	-4.304926000	-1.947583000
H	4.987867000	-5.507185000	-1.821579000
H	6.249702000	-3.728580000	-0.587865000
H	5.100298000	-1.591026000	-0.074160000
H	1.622712000	-2.833902000	-1.882519000
H	2.618658000	-5.038780000	-2.482401000
I	-0.103786000	-0.372418000	-1.462440000
C	2.432310000	2.975273000	-1.375605000
H	1.417132000	2.783393000	-1.723527000
H	2.383143000	3.294248000	-0.333840000
H	2.883431000	3.768138000	-1.975895000
C	5.075399000	1.961324000	-1.004303000
H	5.072552000	2.246302000	0.048486000
H	5.771317000	1.131205000	-1.132685000
H	5.413076000	2.808306000	-1.605387000
C	3.552617000	1.190668000	-3.294754000
H	4.211263000	0.343846000	-3.491935000
H	2.568360000	0.963480000	-3.705709000
H	3.954064000	2.081731000	-3.782201000
C	1.016535000	1.050034000	2.411049000
H	0.090461000	1.063125000	1.835854000
H	0.787154000	0.946798000	3.473667000
H	1.538040000	1.993333000	2.244222000
C	1.137856000	-1.797892000	2.397186000
H	0.204805000	-1.870870000	1.837872000
H	1.728303000	-2.693522000	2.199367000
H	0.920691000	-1.737068000	3.465643000
C	3.490686000	-0.277308000	2.950712000
H	3.177945000	-0.311661000	3.996439000
H	4.144407000	-1.126091000	2.744858000
H	4.055038000	0.639427000	2.774651000

Table S5. DFT calculated Cartesian coordinates of geometry optimized $^3[\text{CuI}(\text{PMe}_3)_2(\text{py})]$

Cu	2.502306000	0.346267000	0.100329000
P	3.421815000	1.509513000	-1.658036000
P	2.211521000	-0.556726000	2.193052000
C	3.749044000	-3.866518000	-1.856235000
C	4.692085000	-3.116076000	-1.103643000
C	4.344658000	-1.923087000	-0.536665000
N	3.075383000	-1.381553000	-0.655464000
C	2.148171000	-2.107453000	-1.392427000
C	2.451034000	-3.304455000	-1.977828000
H	4.006483000	-4.810753000	-2.313360000
H	5.705483000	-3.475757000	-0.967390000
H	5.056209000	-1.341127000	0.033401000
H	1.162610000	-1.667815000	-1.479702000
H	1.676097000	-3.814477000	-2.538513000
I	0.301055000	1.874628000	0.226014000
C	2.462671000	1.155123000	-3.164165000
H	2.458093000	0.078737000	-3.337222000
H	1.436860000	1.494109000	-3.023150000
H	2.902019000	1.662352000	-4.024644000
C	3.488700000	3.329212000	-1.572203000
H	2.484560000	3.716326000	-1.405535000
H	4.123007000	3.624862000	-0.736204000
H	3.894202000	3.739372000	-2.498925000
C	5.121623000	1.023544000	-2.098438000
H	5.794234000	1.263220000	-1.274798000
H	5.154501000	-0.050401000	-2.276707000
H	5.443969000	1.555550000	-2.995281000
C	1.794812000	0.543995000	3.583920000
H	0.896903000	1.108292000	3.337170000
H	1.629390000	-0.036843000	4.493050000
H	2.616072000	1.243209000	3.743531000
C	0.843031000	-1.754801000	2.119599000
H	-0.074485000	-1.232694000	1.849805000
H	1.064695000	-2.497314000	1.352723000
H	0.713273000	-2.249758000	3.083266000
C	3.621223000	-1.531853000	2.809102000
H	3.365968000	-2.005534000	3.758697000
H	3.878781000	-2.295665000	2.076644000
H	4.481431000	-0.877285000	2.951308000

Table S6. IR absorption bands of 1 and 2 in cm⁻¹ (w = weak, m = medium, s = strong, vs = very strong)

1		2	
3047	w	3056/3032	w
1591	w	1583	w
1478	m	1490/1479	w/m
1434	s	1434/1411	s/w
1308	w	1310	w
1214	w	1222/1203	w/w
1185	w	1182	w
1165	w	1157	w
1149	w		
1092	m	1091	m
1067	w	1079/1069	w/w
1032	w		
1026	w	1027	w
1001	m	1010/998/988/850/825	w/w/m/w/m
754	s	750	s
742	s	740	s
692	vs	694	vs
619	w	600	w
527	w		
517	s	514	s
508	s	504	s
500	vs	496	vs
492	s		

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